A blood-based multi-omic landscape for the molecular characterization of kidney stone disease.

PMID: 38623715 | Indexed in PubMed

Kidney stone disease (KSD, also named renal calculi, nephrolithiasis, or urolithiasis) is a common urological disease entailing the formation of minerals and salts that form inside the urinary tract, frequently caused by diabetes, high blood pressure, hypertension, and monogenetic components in most patients. 10% of adults worldwide are affected by KSD, which continues to be highly prevalent and with increasing incidence. For the identification of novel therapeutic targets in KSD, we adopted high-throughput sequencing and mass spectrometry (MS) techniques in this study and carried out an integrative analysis of exosome proteomic data and DNA methylation data from blood samples of normal and KSD individuals. Our research delineated the profiling of exosomal proteins and DNA methylation in both healthy individuals and those afflicted with KSD, finding that the overexpressed proteins and the demethylated genes in KSD samples are associated with immune responses. The consistency of the results in proteomics and epigenetics supports the feasibility of the comprehensive strategy. Our insights into the molecular landscape of KSD pave the way for a deeper understanding of its pathogenic mechanism, providing an opportunity for more precise diagnosis and targeted treatment strategies for KSD.

Genotyping, sequencing and analysis of 140,000 adults from Mexico City.

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The Mexico City Prospective Study is a prospective cohort of more than 150,000 adults recruited two decades ago from the urban districts of Coyoacán and Iztapalapa in Mexico City1. Here we generated genotype and exome-sequencing data for all individuals and whole-genome sequencing data for 9,950 selected individuals. We describe high levels of relatedness and substantial heterogeneity in ancestry composition across individuals. Most sequenced individuals had admixed Indigenous American, European and African ancestry, with extensive admixture from Indigenous populations in central, southern and southeastern Mexico. Indigenous Mexican segments of the genome had lower levels of coding variation but an excess of homozygous loss-of-function variants compared with segments of African and European origin. We estimated ancestry-specific allele frequencies at 142 million genomic variants, with an effective sample size of 91,856 for Indigenous Mexican ancestry at exome variants, all available through a public browser. Using whole-genome sequencing, we developed an imputation reference panel that outperforms existing panels at common variants in individuals with high proportions of central, southern and southeastern Indigenous Mexican ancestry. Our work illustrates the value of genetic studies in diverse populations and provides foundational imputation and allele frequency resources for future genetic studies in Mexico and in the United States, where the Hispanic/Latino population is predominantly of Mexican descent.

Loss of the tumour suppressor LKB1/STK11 uncovers a leptin-mediated sensitivity mechanism to mitochondrial uncouplers for targeted cancer therapy.

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Non-small cell lung cancer (NSCLC) constitutes one of the deadliest and most common malignancies. The LKB1/STK11 tumour suppressor is mutated in ~30% of NSCLCs, typically lung adenocarcinomas (LUAD). We implemented zebrafish and human lung organoids as synergistic platforms to pre-clinically screen for metabolic compounds selectively targeting LKB1-deficient tumours. Interestingly, two kinase inhibitors, Piceatannol and Tyrphostin 23, appeared to exert synthetic lethality with LKB1 mutations. Although LKB1 loss alone accelerates energy expenditure, unexpectedly we find that it additionally alters regulation of the key energy homeostasis maintenance player leptin (LEP), further increasing the energetic burden and exposing a vulnerable point; acquired sensitivity to the identified compounds. We show that compound treatment stabilises Hypoxia-inducible factor 1-alpha (HIF1A) by antagonising Von Hippel-Lindau (VHL)-mediated HIF1A ubiquitination, driving LEP hyperactivation. Importantly, we demonstrate that sensitivity to piceatannol/tyrphostin 23 epistatically relies on a HIF1A-LEP-Uncoupling Protein 2 (UCP2) signaling axis lowering cellular energy beyond survival, in already challenged LKB1-deficient cells. Thus, we uncover a pivotal metabolic vulnerability of LKB1-deficient tumours, which may be therapeutically exploited using our identified compounds as mitochondrial uncouplers.

Assessing the impact of a new medical toxicology service on the treatment of paracetamol overdose at a large tertiary care hospital.

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Paracetamol overdose is the most common cause of acute liver failure in the United States. Administration of acetylcysteine is the standard of care for this intoxication. Laboratory values and clinical criteria are used to guide treatment duration, but decision-making is nuanced and often complex and difficult. The purpose of this study was to evaluate the effect of the introduction of a medical toxicology service on the rate of errors in the management of paracetamol overdose. This was a single center, retrospective, cohort evaluation. Patients with suspected paracetamol overdose were divided into two groups: those attending in the 1 year period before and those in the 1 year after the introduction of the medical toxicology service. The primary outcome was the frequency of deviations from the established management of paracetamol intoxication, using international guidelines as a reference. Fifty-four patients were eligible for the study (20 pre-toxicology-service, 34 post-toxicology-service). The frequency of incorrect therapeutic decisions was significantly lower in the post-toxicology service implementation versus the pre-implementation group ($P = 0.005$). Our study suggests that a medical toxicology service reduces the incidence of management errors, including the number of missed acetylcysteine doses in patients with paracetamol overdose. The limitations include the retrospective study design and that the study was conducted at a single center, which may limit generalizability. The implementation of a medical toxicology service was associated with a decrease in the number of errors in the management of paracetamol overdose.

Early treatment with N-acetylcysteine reduces hepatotoxicity in acute acetaminophen poisoning.

PMID: 37247196 | Indexed in PubMed

During the outbreak of COVID19 measures taken to contain the spread of the virus have influenced the mental well-being of adults and adolescents. Acetaminophen overdose is the major cause of drug intoxication among children and adolescents. We reported a case of a 15-year- old girl referred to our Emergency Department 3 hours after ingestion of 10 g of paracetamol for suicidal purposes. She promptly started the administration of intravenous N-acetylcysteine (NAC) and the patient was discharged after 5 days of hospitalization in good clinical condition and with neuropsychiatric follow-up. Our case shows that the timing of the intravenous NAC administration is considered the most important factor in the prevention of acetaminophen-induced hepatic failure, despite high serum levels after acetaminophen ingestion.

[Simplified N-acetylcystein treatment after paracetamol overdose - new recommendations from Swedish Poisons Information Centre].

PMID: 31688944 | Indexed in PubMed

Since the late 1970s N-acetylcystein has been used as an antidote after paracetamol intoxication. The treatment is traditionally given as three consecutive infusions for 20 hours and 15 minutes. The total dose given is 300 mg/kg. Half of this amount is given as a bolus during the first 15 minutes of treatment. This regime has proven very efficient in avoiding liver injury. However, side effects, caused by histamine release, are common (10-15%). Symptoms as flush, urticaria and, in rare cases, bronchospasm, angioedema and circulatory shock typically appear during the bolus dose and may lead to interrupted and inadequate treatment. In addition, the regime is complicated leading to a risk of administration errors. During the last years several publications have described the use of a model with two infusions instead of three. The first and the second infusions are merged and given over four hours. The third infusion and the total dose are left unchanged. This modified regime has been shown to reduce side effects and seems not to increase the risk of liver injury. As of November 1, 2019, the Swedish Poisons Information Centre will change its recommendations to the new two-infusion protocol.

Recommendations for the paracetamol treatment nomogram and side effects of N-acetylcysteine.

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Treatment of paracetamol intoxication consists of administration of N-acetylcysteine, preferably shortly after paracetamol ingestion. In most countries, the decision to treat patients with N-acetylcysteine depends on the paracetamol plasma concentration. In the literature, different arguments are given regarding when to treat paracetamol overdose. Some authors do not recommend treatment with N-acetylcysteine at low paracetamol plasma concentrations since unnecessary adverse effects may be induced. But no treatment with N-acetylcysteine at higher paracetamol plasma concentrations may lead to unnecessary severe morbidity and mortality. In this review, we provide an overview on the severity and prevalence of adverse side effects after N-acetylcysteine administration and the consequences these side effects may have for the treatment of paracetamol intoxication. The final conclusion is to continue using the guidelines of the Dutch National Poisons Information Centre for N-acetylcysteine administration in paracetamol intoxication.

Paracetamol intoxication and N-acetyl-cysteine treatment.

PMID: 7491847 | Indexed in PubMed

N-acetylcysteine is a good and reasonable specific antidotum in case of paracetamol intoxication. It is very active when administered within eight hours after intoxication. It can be used as well as in intravenous as peroral administration. There are few side effects.

Criteria for acetylcysteine treatment and clinical outcomes after paracetamol poisoning.

PMID: 22697593 | Indexed in PubMed

Acetylcysteine is an effective antidote for paracetamol (acetaminophen) poisoning, but different treatment criteria exist internationally. In the UK, acetylcysteine is indicated by paracetamol concentrations higher than the Prescott nomogram or higher than 50% of the nomogram in patients with increased susceptibility to liver toxicity. In the USA, a single '150-line' nomogram has been used that removes the need for additional clinical risk assessment. The latter approach has recently been adopted in Australia, New Zealand and elsewhere. Few data exist to allow direct comparison of these different international approaches. An existing database of 1191 patients admitted to hospital after paracetamol overdose identified that the 4-h equivalent paracetamol concentration was: ≥200 mg/l in 163 patients (15.6%; 95% CI: 13.3-18.2%), ≥150 mg/l in 264 (24.3%; 95% CI: 21.5-27.5%) and ≥100 mg/l in 426 patients (39.3%; 95% CI: 35.6-43.2%), and acute liver injury occurred in 3.7% (95% CI: 1.4-8.0%), 2.3% (95% CI: 0.8-5.0%) and 1.9% (95% CI: 0.8-3.7%), respectively. The different indications for acetylcysteine used by the UK and USA would result in similar numbers of patients treated, although the criteria would define patients with different characteristics and patterns of overdose. The relative merit of these different international approaches to acetylcysteine administration is considered in this article.

[Paracetamol poisoning].

PMID: 7753607 | Indexed in PubMed

Administration of paracetamol (acetaminophen) has analgetic and antipyretic effect. After trauma paracetamol has an anti-inflammatory activity. It was presumed that paracetamol in therapeutic doses had fewer and more acceptable side-effects than other analgetic drugs such as acetylsalicylic acid and NSAID-drugs. However, in toxic concentrations, paracetamol is more life-threatening. The toxic effects of paracetamol most often occur in the liver and kidneys. Phosphate and lactate turn-over can also be impaired. Paracetamol poisoning can induce temporary liver disfunction or even irreversible liver failure with liver transplantation as the only therapeutic possibility. Chronic alcoholics are especially at risk, as liver damage may occur following paracetamol even in recommended doses. When intoxication with paracetamol is presumed, administration of N-acetylcysteine is vital. N-acetylcysteine therapy should be initiated not later than 15 hours after paracetamol intake. Moreover, the antitoxic effect has been observed even when N-acetylcysteine therapy is initiated 24-36 hours after presumed paracetamol intake. Measures of preventing further absorption of paracetamol from the gastrointestinal tract should be taken. Activated charcoal should be given if less than two hours have passed since paracetamol intake. Between two and four hours following paracetamol intake gastric lavage should be performed. During the last 10 years the incidence of paracetamol self-poisoning has increased, but death following paracetamol poisoning is relatively constant at around nine per year in Denmark. It is suggested that the incidence of serious cases of paracetamol poisoning could be reduced by simple measures. Special attention should be paid to the risk-group of chronic alcoholics.